# SIMATS ENGINEERING
## ASSESSMENT TOOL 2

**COURSE CODE: ITA0605** **COURSE NAME: MACHINE LEARNING**

**COURSE OUTCOME COVERED**

***CO1:*** *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

*Assessment Tool Weightage: 20%*

**QUESTIONS: 15** **Total Marks: 25**

Annexure A

**CONCEPT MAPPING**

***Code of Conduct:***

*I Thamizharasan S (192424087) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***
THAMIZHARASAN S

Annexure A

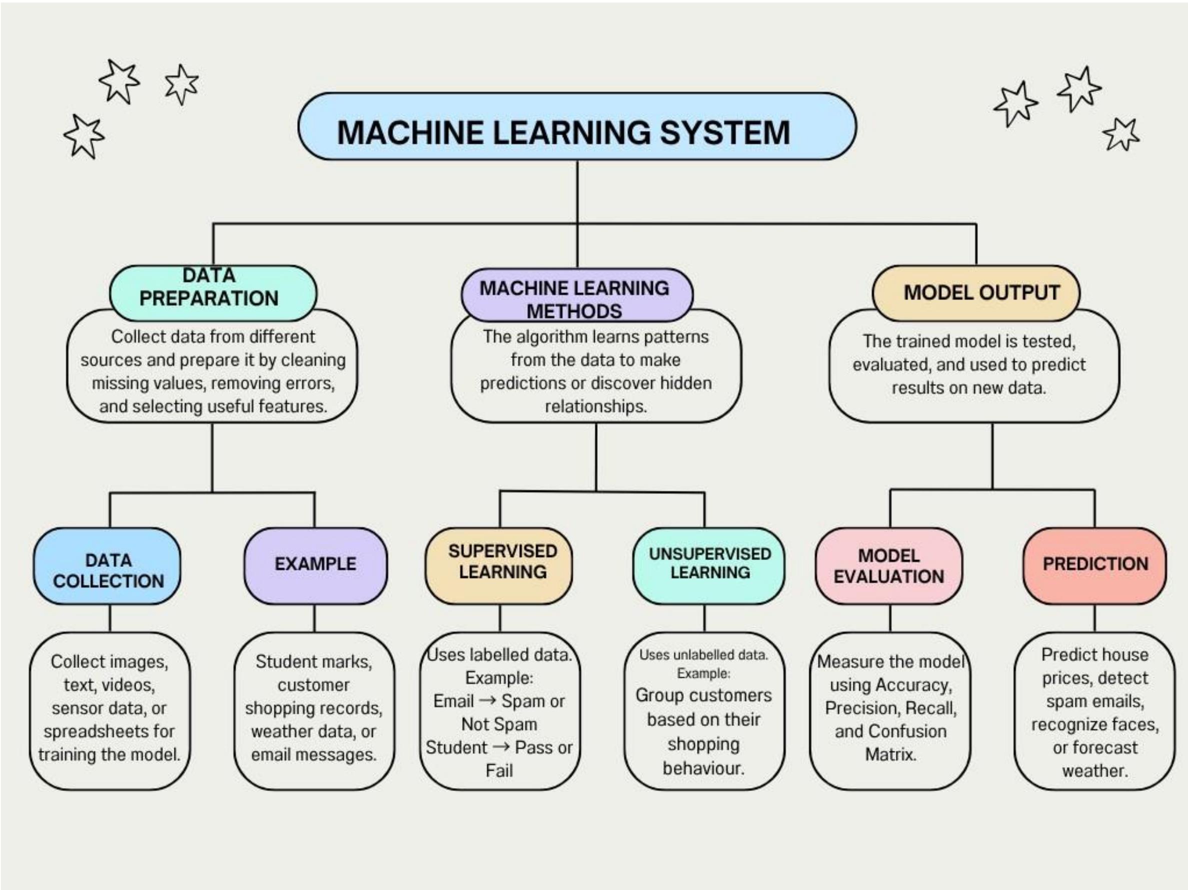
**Assessment Tasks**

**Concept Mapping Tasks**

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

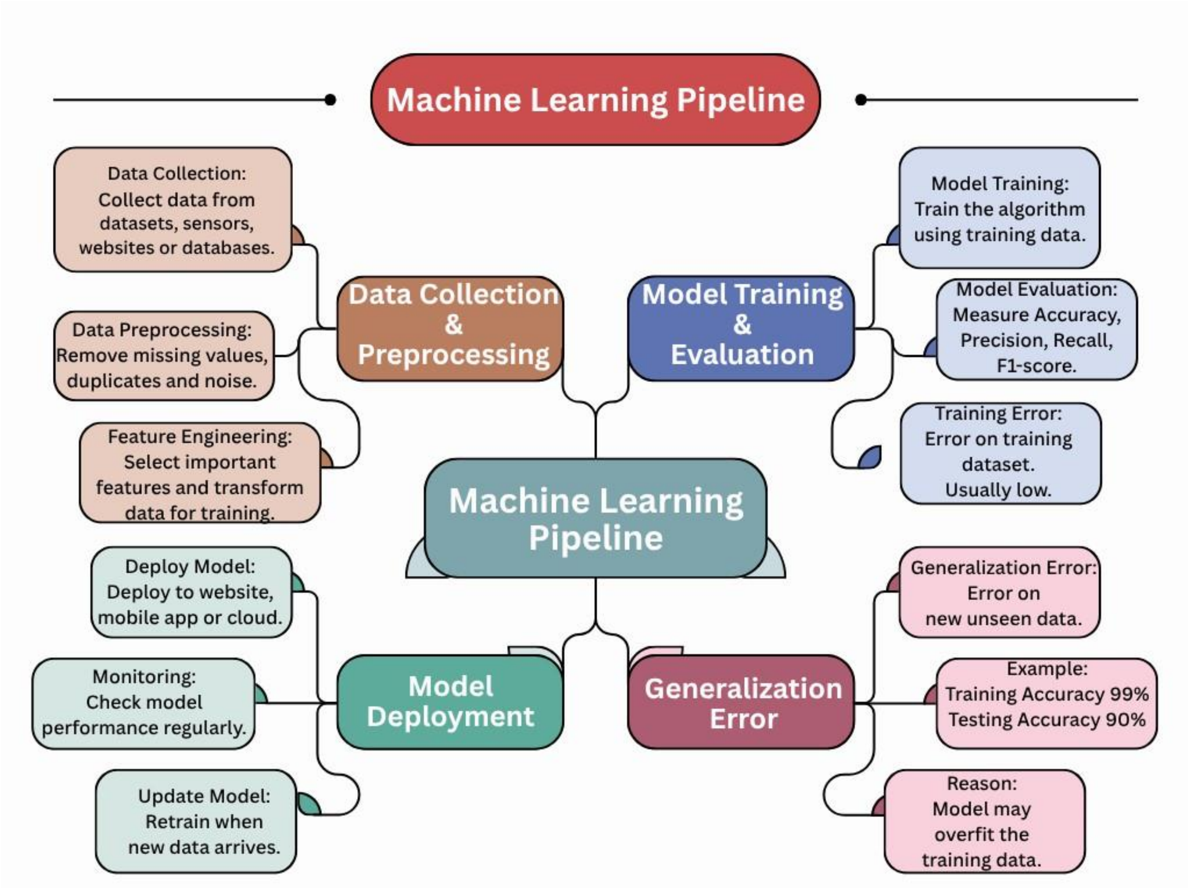
Concept Map 1:

With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme. (5 Marks)



Concept Map 2:

Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples. (5 Marks)

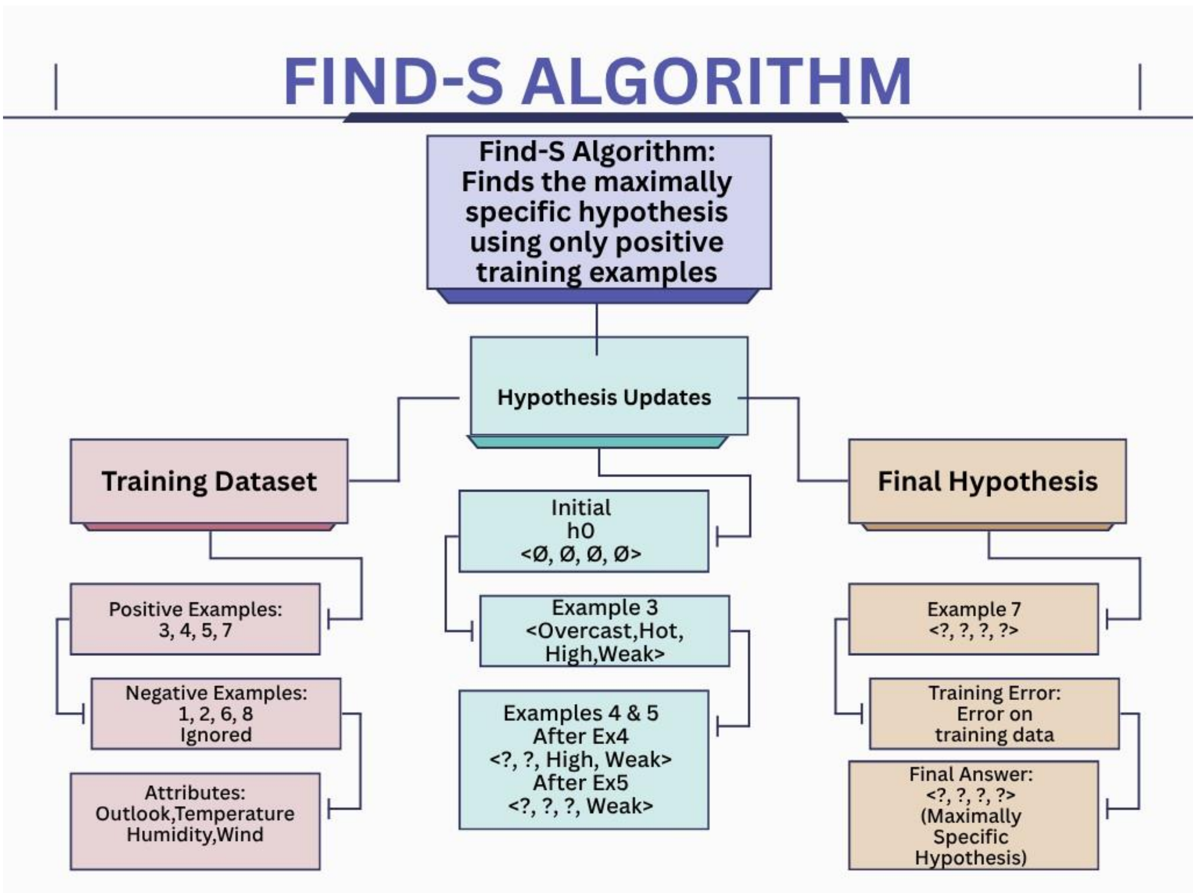


Concept Map 3:

In the context of Find-S algorithm, consider the following dataset used to determine whether a day is suitable for playing tennis based on attributes such as Outlook, Temperature, Humidity, and Wind. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show the step-by-step updates of the hypothesis after each positive training example and ignore all negative examples. Finally, provide the final hypothesis. (5 Marks)

**Dataset:**

| Example | Outlook | Temperature | Humidity | Wind | Play Tennis |
|---|---|---|---|---|---|
| 1 | Sunny | Hot | High | Weak | No |
| 2 | Sunny | Hot | High | Strong | No |
| 3 | Overcast | Hot | High | Weak | Yes |
| 4 | Rainy | Mild | High | Weak | Yes |
| 5 | Rainy | Cool | Normal | Weak | Yes |
| 6 | Rainy | Cool | Normal | Strong | No |
| 7 | Overcast | Mild | Normal | Strong | Yes |
| 8 | Sunny | Mild | High | Weak | No |



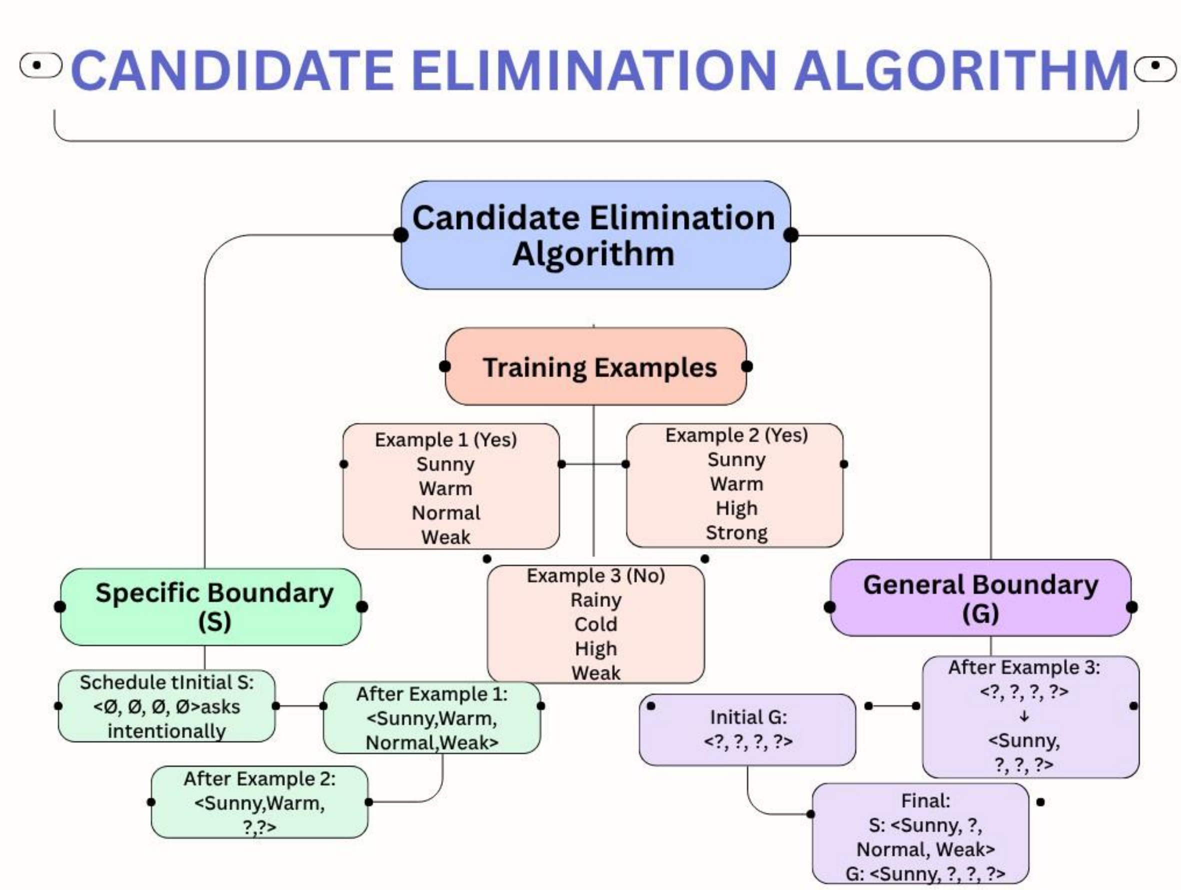
Concept Map 4:

In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a person will enjoy a sport based on attributes such as Weather, Temperature, Humidity, and Wind. Using the Candidate Elimination algorithm, determine the version space by computing both the specific (S) and general (G) boundaries. Show the step-by-step updates after processing each training example.

(5 Marks)

**Dataset:**

| Example | Weather | Temperature | Humidity | Wind | Enjoy Sport |
|---|---|---|---|---|---|
| 1 | Sunny | Warm | Normal | Weak | Yes |
| 2 | Sunny | Warm | High | Strong | Yes |
| 3 | Rainy | Cold | High | Weak | No |
| 4 | Sunny | Cold | Normal | Weak | Yes |
| 5 | Rainy | Warm | Normal | Strong | No |



| Criteria | Weightage | Level 4 - Exemplary | Level 3 - Proficient | Level 2 - Developing | Level 1 - Beginning |
|---|---|---|---|---|---|
| Concept Identification | 25% | All major concepts are accurately identified | Most concepts are identified correctly | Limited concepts are identified | Essential concepts are missing |
| Relationship Mapping | 25% | Links between concepts are accurate and meaningful | Most links are correct | Some links are weak or unclear | Relationships are mostly incorrect |
| Hierarchy and Organization | 20% | Excellent structure with clear hierarchy | Mostly organized with minor issues | Basic structure with weak hierarchy | No logical organization |
| Technical Relevance | 20% | Map strongly reflects cloud course content | Mostly aligned to topic | Partially aligned to topic | Weak alignment to topic |
| Visual Clarity | 10% | Neat, readable, and easy to interpret | Generally readable | Somewhat cluttered | Poorly presented |

**Permitted Tools :**
**draw.io / diagrams.net, Lucidchart, MindMeister, Miro, Canva, PowerPoint, Google Drawings, or equivalent diagramming tool**

## Procedure for Preparing the Digital Concept Map

1. Use the following procedure to prepare the diagram and upload evidence.
2. Students who already know another diagramming tool may use it, but the exported evidence and source file must still be submitted.
3. Open https://app.diagrams.net/ or the desktop version of draw.io / diagrams.net.
4. Choose Device, Google Drive, or OneDrive as the save location according to availability.
5. Create a blank diagram and name the file as ITAO6_AT2_RegisterNumber_Concept_Map.drawio.
6. Place the concept map title at the center or as a clearly visible anchor node.
7. Add major cloud nodes: cloud definition, cloud evolution, cloud architecture, service models, deployment models, virtualization, web services, SOAP, REST, APIs, elasticity, scalability, resource pooling, measured service, and emerging paradigms.
8. Connect nodes with labeled arrows such as uses, depends on, enables, scales through, communicates through, belongs to, supports, provides, consumes, or controls.
9. Use colors meaningfully: one color for service models, one for cloud characteristics, one for API/web-service concepts, and one for concept map title-specific elements.
10. Export the final diagram as PNG or PDF. Keep the editable .drawio source file also.
11. Paste the exported diagram image or screenshot in this DOCX under Task 1.
12. Upload the DOCX, exported PNG/PDF, and .drawio file to Google Classroom.